



**Bharati Vidyapeeth's
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SKILLSYNC

An AI//ML-Based Learner Recommendation System for Skill-Gap Analysis,
Personalized Roadmaps and Verified Peer/Mentor Matching

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TABLE OF CONTENTS

- Introduction
- Problem Statement
- Literature Survey
- Research Gap
- Objectives
- Proposed Methodology
- Performance Comparison of Existing Methodology
- Current Progress / Expected Outcomes
- References in IEEE Format



INTRODUCTION

THE PROBLEM

Learners today have access to thousands of courses, tutorials and online communities, but often face three major problems:

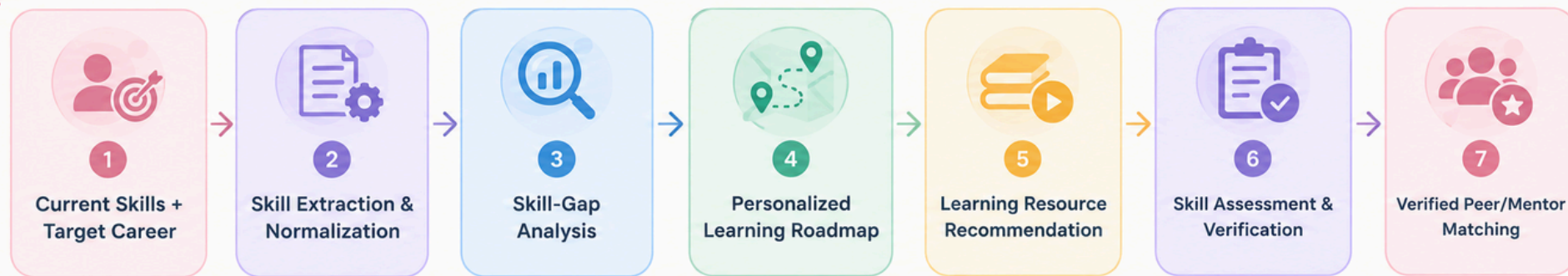
- **What am I missing?** — No precise identification of skills required for a target career.
- **What should I learn first?** — Available resources are not always organized into a personalized learning path.
- **Whom should I ask?** — Online communities recommend people based on popularity or activity, not verified possession of a specific skill.

EXAMPLE

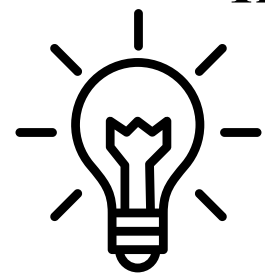
A student wants to become a Data Scientist but knows only Python and basic Statistics. SkillSync identifies missing skills like SQL, ML, and Data Visualization, then creates a learning roadmap and connects them with verified mentors/peers.

INTRODUCTION : OUR SOLUTION

SKILLSYNC — FROM SKILL GAP TO SKILL GROWTH



KEY IDEA



Instead of solving only “What should I learn?”, SkillSync combines:
Gap Diagnosis + Roadmap + Resources + Skill Verification + Human Guidance
into one explainable recommendation pipeline.



DETAILED LITERATURE SURVEY-I

Skill Gap & Learning Path Recommendation

Research	Approach	Key Contribution	Limitation
Braun et al. (2025)	Qualitative skill-gap analysis	Establishes skill-gap management as a continuous process	No computational skill-gap detection or recommender system
Shi et al. (2020)	Multidimensional Knowledge Graph	Generates prerequisite-aware learning paths	KG construction is complex; no peer/mentor matching
Haldar et al. (2025)	Graph Reinforcement Learning	Personalizes learning paths using sequential decisions	Requires significant interaction data and is less explainable
Zhang et al. (2023)	Multi-context Knowledge Graph	Uses learner, concept and community context	Data-heavy and focused mainly on path recommendation

TAKEAWAY- Existing research has made learning paths increasingly personalized and context-aware, but primarily focuses on path generation, not the complete learner journey.



DETAILED LITERATURE SURVEY-II

Personalization, Behaviour & LLM-Based Approaches

Research	Approach	Contribution	Limitation
Fan et al. (2026)	Forgetting factor + Knowledge Graph	Considers knowledge retention while generating paths	Requires additional learner-history modelling
Yang et al. (2026)	Goal-oriented learning + forgetting curve	Aligns learning paths with goals and retention	Focuses on path generation rather than human support
Ma (2025)	ML-based behaviour analysis	Uses learner behaviour for personalized recommendation	Depends on behavioural data; does not verify mentors
Multimodal Learning Path (2025)	Behaviour + Sentiment Analysis	Adds emotional/feedback signals to personalization	Data-hungry and adds privacy/modelling complexity
Deshmukh et al. (2026)	LLM + Sequential Recommendation	Generates flexible personalized learning sequences	Cost, latency, hallucination and explainability concerns

TAKEAWAY- These approaches improve personalization, but introduce complexity and still do not connect the learner to verified providers of the exact missing skills.



RESEARCH GAP

EXISTING SYSTEMS generally solves one or two parts:

**Skill Gap
Analysis**

**Learning
Path Systems**

**Resource
Recommenders**

**Peer/Tutor
Systems**

**Assessment
Systems**

There is no single lightweight and explainable pipeline that combines:
**Current Skills + Target Career → Skill-Gap Analysis → Personalized Roadmap +
Resources → Skill Verification → Skill-Specific Verified Peer/Mentor Matching**

OUR CONTRIBUTION:

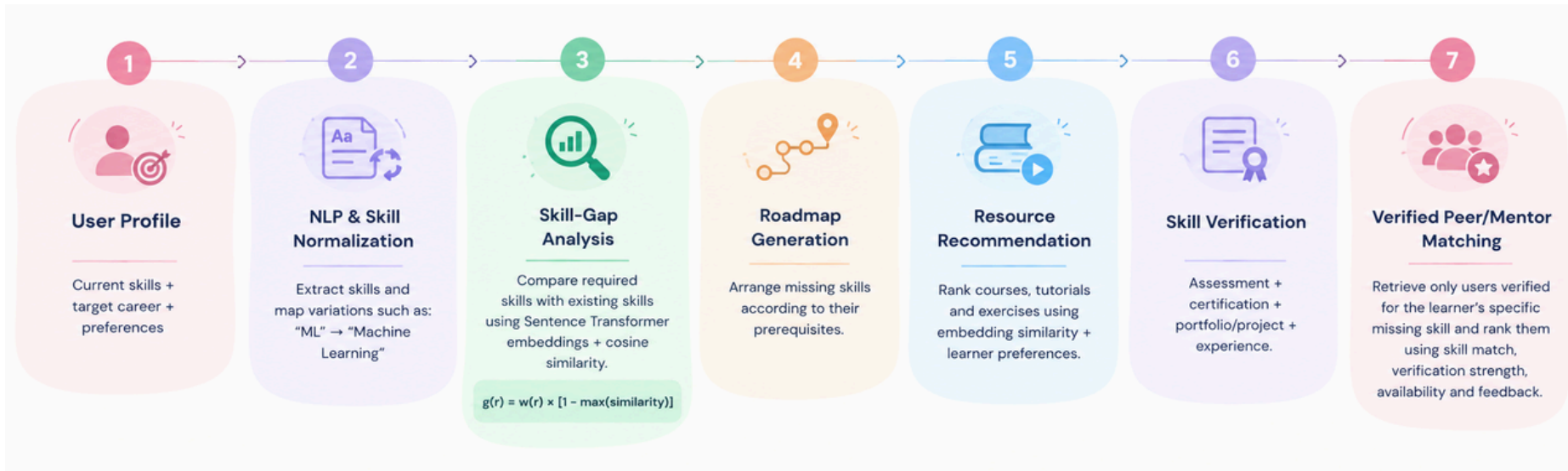
A learner is matched only with people verified for the particular skill they are missing, rather than generic or self-declared expertise.



OBJECTIVES

- **O1.** To design a learner profile module that accepts current skills as free text or a structured list and a target career/learning goal, and to extract and normalize skills using NLP and embedding techniques.
- **O2.** To construct a curated mapping from target goals to required-skill sets, and to compute an interpretable skill gap using embedding-based similarity and a per-skill gap score.
- **O3.** To generate a prerequisite-ordered personalized learning roadmap over the identified gap skills.
- **O4.** To recommend relevant, ranked learning resources for each gap skill using content/embedding features and cosine similarity.
- **O5.** To design and implement a skill-verification mechanism whereby users become Verified Peers/Verified Mentors/Verified Skill Providers for specific skills via platform skill assessments, supported by certifications, portfolio/projects and experience.
- **O6.** To retrieve and rank Verified Peers/Verified Mentors matched to the learner's specific missing skills, incorporating a diversity/fairness consideration to avoid over-recommending a few providers.
- **O7.** To integrate the above into one explainable, lightweight web application with a React frontend and a Python (Flask/FastAPI) backend, exposing the rationale for each recommendation.

PROPOSED METHODOLOGY-PIPELINE



- SkillSync starts by analyzing the learner's profile, skills, and career goals.
- It identifies skill gaps and generates a personalized, prerequisite-based learning roadmap with relevant resources.
- The system then verifies skills through assessments, certifications, projects, and experience.
- Finally, it connects learners with verified peers or mentors who possess the required skills.



PERFORMANCE COMPARSION OF EXISTING METHODOLOGY

Methodology	Skill Gap	Roadmap	Resources	Verified Peer/Mentor
Qualitative Skill-Gap Analysis	✓	✗	✗	✗
Knowledge Graph	✗	✓	Partial	✗
Graph Reinforcement Learning	✗	✓	✗	✗
Behaviour/Sentiment Models	✗	✓	✓	✗
LLM-Based Recommendation	Partial	✓	Partial	✗
Peer Recommendation	✗	✗	✗	Partial
ML + Automated Assessment	✗	✗	✗	✓
Diversity-Aware Tutoring	✗	✗	✗	Partial
SkillSync	✓	✓	✓	✓



CURRENT PROGRESS

Literature Review Completed

Studied existing research on skill-gap analysis, personalized learning paths, recommendation systems, and peer/mentor matching.

Frontend Developed

Designed the initial SkillSync interface and user flow for the proposed system.

System Planning Completed

Finalized the overall workflow, major modules, and approach to be followed for the backend and ML implementation.

EXPECTED OUTCOME

- Accurate Skill-Gap Identification for the learner's target career.
- Personalized Learning Roadmap with prerequisite-based learning steps.
- Relevant Resource Recommendations for each identified skill gap.
- Verified Peer/Mentor Matching based on required skills, verification, and relevance.



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